

KHOI ANH NGUYEN

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GIS Portfolio  [Portfolio.html](#) • GitHub  [khoi9034/ArcPyScripts](#)

University of North Carolina at Chapel Hill • Aug 2022 – May 2026

B.A Geography | Minor GIS

TECHNICAL SKILLS

GIS Software: ArcGIS Pro, ArcGIS Online, QGIS

Programming: ArcPy (Python), ModelBuilder, GitHub, JavaScript, HTML, CSS, R (Spatial)

Data Management: Geodatabases, Shapefiles, Projections, SQL (Basic)

Finance & Analysis: DCF valuations, Excel (Pivot Tables, VLOOKUP, etc)

GIS Methods: Geoprocessing, Spatial Analysis, Raster/Vector Analysis, Map Production

ARCPY AUTOMATION & SPATIAL ANALYSIS

- Automated buffers, raster/vector conversions, batch geoprocessing, and feature class exports.
- Developed workflows for LISA, Getis-Ord Gi* hotspot analysis, and Near Table proximity calculations.
- Integrated external data sources (Google Places API, CSVs) for reproducible spatial analysis.
- Validated projections, CRS, and metadata to ensure spatial accuracy.
- Processed DEM mosaics, calculated population densities, and created raster/vector layers for topography and accessibility.

GIS & RESEARCH PROJECTS

- Clustering in Tokyo: Anime Store Spatial Patterns — Triangulated LISA, Near Tables, and Gi* hotspot analysis; automated full ArcPy workflow; identified population clusters and cultural hotspots (Akihabara, Ikebukuro, Shibuya).
 - Retail Hotspot & Spatial Clustering Analysis — Conducted LISA and Gi* hotspot analysis; automated geocoding, data cleaning, and map export.
 - Accessibility to Long-Term Care Facilities — Analyzed elderly access using ArcPy automation and GIS overlay; identified underserved areas for planning.
 - Diversity Index Analysis — Calculated Simpson's Diversity Index across Orange County; automated multi-year census processing.
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WORK EXPERIENCE

Financial Analyst and Investor (Personal Portfolio) • Dec 2021 – Present

- Conduct fundamental equity research using financial statements and cash flow analysis.
- Build DCF models and Excel-based analyses using pivot tables and VLOOKUP/XLOOKUP.
- Investigate customer demand, market size (TAM, SAM, SOM), and customer adoption trends.

Chef Assistant, UNC Campus Dining • 2022 – 2023

- Supported high-volume kitchen operations serving 500+ students per shift.
- Maintained inventory, restocked supplies, and prepared food efficiently.
- Performed accurately and reliably under fast-paced pressure.